Verification scope of the Chua tangency argument

Interval certificate · partial formal verification · coexistence hypotheses

1 NUMERICAL CERTIFICATE

Chua return map
80 and 100 digits: all 15 gates pass
Two shifted boxes: both rejected
135 transverse flights per connection

2 LEAN-CHECKED IMPLICATION

Arbitrary continuous gap g(a, b)
Four face signs + actual derivatives
⇒ an interior $g = 0$ and $g_a = 0$
Also: ten exact rational guards

3 CLASSICAL THEOREM

Newhouse–Robinson
Applies within a fixed parameter family
Can give infinitely many sinks
at a single parameter value

SCOPE OF THE FORMALIZATION

The interval evaluator and exact invariant manifolds
are not encoded in Lean. The general theorem checks
the logical implication once its hypotheses hold.
The Chua-specific hypotheses are checked separately.

LOCALIZATION: ANALYTICAL PROOF

Local returns + the finite excursion stay inside
the domain where the extension agrees with Chua.
The parameter theorem gives infinitely many sinks.
This argument is not formalized in Lean.

Existence theorem Convexity is not needed for existence of the stationary zero.

Dashed arrows indicate connections outside Lean. See the companion basin and coexistence report for localization.